

DATE:-

INTEGRATOR AND DIFFERENTIATOR USING OP-AMP

Expt NO:

AIM:-

To design and setup integrator and differentiator circuit using op-amp.

APPARATUS REQUIRED:-

Power supply, CRO, function generator, breadboard, op-amp, capacitors and resistors.

THEORY:-

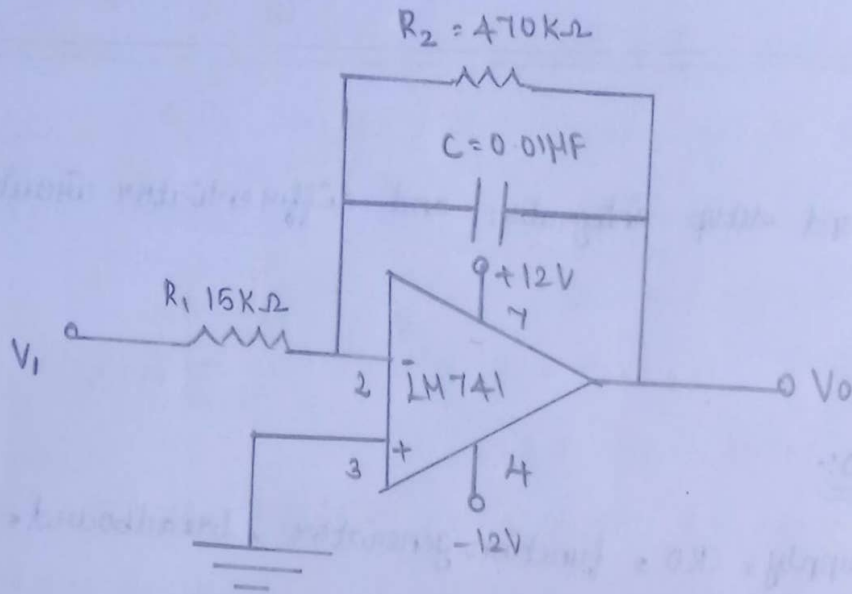
INTEGRATOR:-

Refer to the figure 1. The circuit performs the integration of the input waveform. The output voltage V_o can be expressed by $V_o = -\frac{1}{RC} \int V_i dt + k$ where k is the constant of integration which depends on the value of V_o at $t=0$. The peak of the output waveform V_T is given by the expression $V_T = \frac{V_i T}{4RC}$, where T is the time period of the input square wave. Integrators are commonly used in analog computers and wave shaping networks.

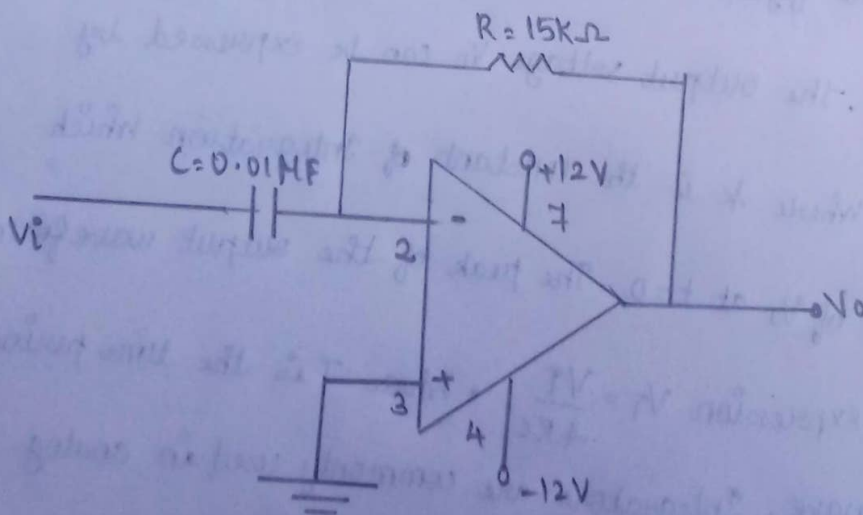
DIFFERENTIATOR:-

If the input resistor of the inverting amplifier is replaced by a capacitor, it forms an inverting differentiator. The output of the circuit is the derivative of the input. Gain of the differentiator

DIAGRAM:-



CIRCUIT DIAGRAM OF INTEGRATOR



CIRCUIT DIAGRAM OF DIFFERENTIATOR

increases with increases in frequency, which makes the circuit unstable. This is a drawback of the circuit. The output voltage V_o can be expressed in $V_o = -R_f C \frac{dV_i}{dt}$. Differentiation function has a high pass filter. At high frequency it becomes unstable and breaks into oscillations. Input Impedance decreases with increases in frequency which makes the circuit susceptible to high frequency noise. Both stability and high frequency noise problems can be reduced significantly by additional circuit elements.

DESIGN AND CIRCUIT DIAGRAM:-

Design of Integrator:-

Let the input frequency be 1KHz. The frequency at which the integrator gives unity gain output is given by,

$$f = \frac{1}{2\pi R_1 C}$$

Let $C = 0.01\mu F$, then $R_1 = 15.9 K\Omega$. Use $15 K\Omega$ std.

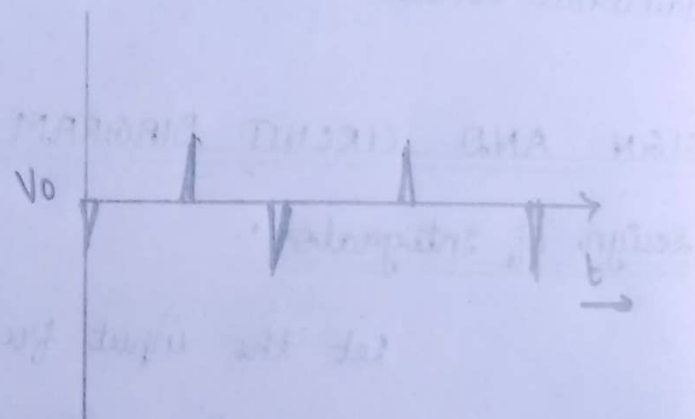
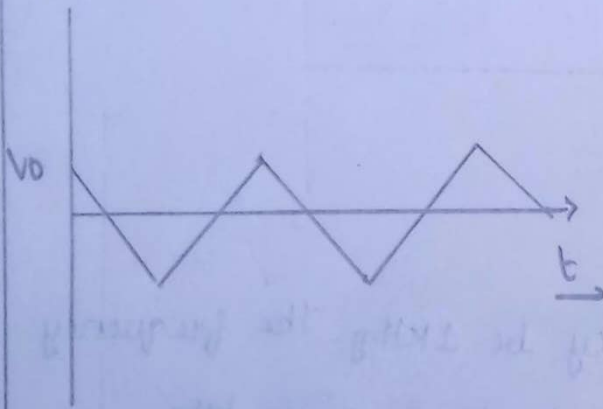
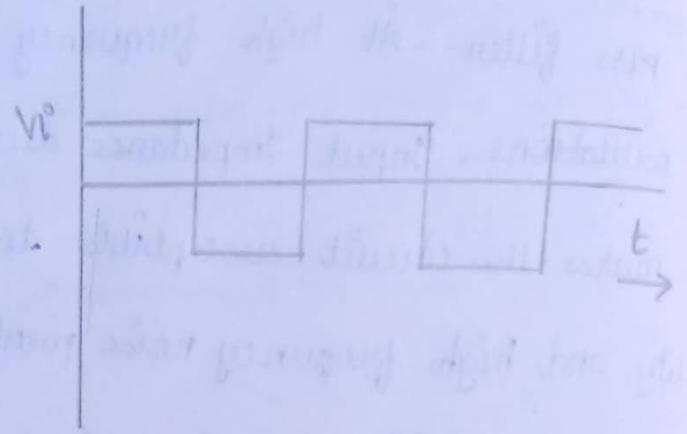
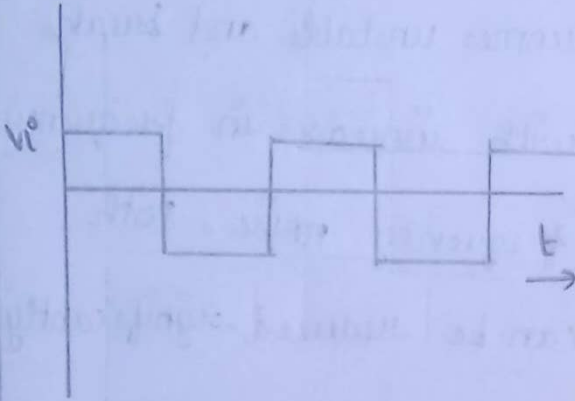
The Resistor R_2 in the Integrator is provided to attenuate low frequency signals particularly input dc offset voltage that may be present. Typically, the value of R_2 is selected 10 times R_1 or more. select the values of R_2 as $170 K\Omega$.

DESIGN OF DIFFERENTIATOR:-

we have, $f = \frac{1}{2\pi RC}$

Let $C = 0.01\mu F$, then $R = 15.9 K\Omega$. Use $15 K\Omega$ std.

WAVEFORMS:-



Integrator Output

Differentiator Output

PROCEDURE:-

INTEGRATOR:-

1. Setup the Integrator circuit as shown in the figure. Give a rectangular wave of $\pm 5V$ (10pp) and 1KHz frequency at the input and observe the input and output simultaneously on CRO.
2. Vary the dc offset voltage of the square wave input and observe the difference in the output waveform.
3. Repeat the experiment by feeding triangular wave and sine wave at the input and observe the output.

DIFFERENTIATOR:-

1. Setup the differentiator circuit as shown in the figure. Give a rectangular wave of $\pm 5V$ (10pp) and 1KHz frequency at the input and output simultaneously on CRO.
2. Repeat the experiment by feeding triangular wave and sine wave at the input and observe the output.

RESULT:-

Thus, the integrator and differentiator circuits were designed and implemented using op-amp, and their output waveforms were obtained and studied successfully.